解方程

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{y}{x^6y^3-x}$$

$$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y}=\frac{1}{y}(x^6y^3-x)$$

$$\Rightarrow \frac{\mathrm{d}x}{x}=\frac{\mathrm{d}y}{y}(x^5y^3-1)$$

令$s=\ln x, t=\ln y$

$$\Rightarrow \mathrm{d}s=\mathrm{d}t(e^{5s+3t}-1)$$

令$5s+3t=u$

$$\Rightarrow \mathrm{d}u=\mathrm{d}t(5e^u-2)$$

$$\Rightarrow \frac{1}{2}\ln(5-2e^{-5s-3t})+C=t$$

$$\Rightarrow \frac{1}{2}\ln(5-\frac{2}{x^5y^3})+C=\ln y$$